THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 8 · Term 3

Student edition

Outcomes: MA-8.4 surface area and volume of pyramids, cones, spheres and hemispheres; MA-8.6 range and standard deviation, frequency polygons, experimental vs theoretical probability.

Lesson notes

- **Pyramids and cones (Weeks 1–4):** volume of a pyramid = ⅓ × base area × height; volume of a cone = ⅓πr²h; curved surface of a cone = πrl (l = slant height). Fill a cone with rice from a cylinder of the same base and height to show the ⅓.

  ▶ **Try it on NovaSchool:** How many cones fill a cylinder? — open **Simulations** on your dashboard.

- **Spheres and hemispheres (Weeks 5–8):** volume of a sphere = 4/3 πr³; surface area = 4πr²; hemisphere volume = 2/3 πr³. Real objects: footballs, domes of mosques.
- **MA-8.6 (Weeks 9–12):** range = highest − lowest; standard deviation measures how spread out data is (find the mean, the squared differences, their mean, then the square root). Frequency polygons (join midpoints). Experimental probability = times it happened ÷ number of trials; it gets closer to the theoretical value with more trials. Class coin and dice experiment.

Workbook pages

Page	Task
1–2	Volume and surface area of 8 solids.
3	Rice experiment record.
4–5	Range and standard deviation of two data sets; frequency polygon.
6	50 coin tosses: experimental vs theoretical.

Question bank

#	Outcome	Question	Your answer
1	MA-8.4	Volume of a pyramid with a 6 cm × 6 cm base and height 10 cm?	360 cm³ / 120 cm³ / 60 cm³ / 180 cm³
2	MA-8.4	Volume of a cone with r = 3 cm, h = 7 cm (π = 22/7)?	198 cm³ / 66 cm³ / 22 cm³ / 132 cm³
3	MA-8.4	Surface area of a sphere with r = 7 cm (π = 22/7)?	154 cm² / 308 cm² / 616 cm² / 1,437 cm²
4	MA-8.4	A cone holds one third of a cylinder with the same base and height. True or false.	True / False

#	Outcome	Question	Your answer
5	MA-8.6	Range of 4, 9, 2, 15, 7?	11 / 13 / 15 / 7
6	MA-8.6	Data 2, 4, 4, 4, 5, 5, 7, 9 has mean 5. Its standard deviation?	1 / 2 / 4 / 5
7	MA-8.6	A coin landed heads 18 times in 40 tosses. Experimental P(heads)?	0.5 / 0.45 / 0.18 / 0.4
8	MA-8.6	With more trials, experimental probability usually gets closer to theoretical probability. True or false.	True / False
9	MA-8.6	A frequency polygon joins the … of each class.	lower limits / midpoints / upper limits / totals